3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
thumu_hemanth_kumar@Hen
  GNU nano 8.7.1                                    practical3.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Parent process started\n");
    printf("Parent PID  : %d\n", getpid());
    printf("Parent PPID : %d\n\n", getppid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("----- Child Process -----\n");
        printf("Child PID  : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child is running...\n");
                                  [ Read 48 lines ]
```

```
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSPractical$ ./practical3
Parent: PID = 1098, Child PID = 1099
Child: PID = 1099, PPID = 1098
```

```
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ ls /proc
1     173  354  80          cpuinfo        filesystems  keys            misc          schedstat      sysvipc
1014  180  400  88          crypto         fs           kmsg            modules       scsi           thread-sel
1015  2    402  acpi        devices        interrupts   kpagecgroup     mounts        self           timer_list
1022  206  427  buddyinfo   diskstats      iomem        kpagecount      mpt           slabinfo       tty
1098  216  47   bus         dma            ioports      kpageflags      mtrr          softirqs       uptime
1099  251  688  cgroups     driver         irq          latency_stats   net           stat           version
1105  271  689  cmdline     dynamic_debug  kallsyms     loadavg         pagetypeinfo  swaps          vmallocinf
171   311  691  config.gz   execdomains    kcore        locks           partitions    sys            vmstat
172   315  7    consoles    fb             key-users    meminfo         pressure      sysrq-trigger  zoneinfo
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$
```